Holomorphic Hopf–Beltrami Maxwell Modes on the Round Three-Sphere

Exact Construction, Conformal Transport, and Penrose Representation

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Abstract

We construct an explicit fixed-polarization family of global null Maxwell modes on the Einstein cylinder $\mathbb{R} \times S^3$. On the oriented unit round three-sphere, let X_1, X_2, X_3 be a global orthonormal Hopf frame and set $X = X_2 + iX_3$. For every homogeneous holomorphic polynomial $p(z_1, z_2)$ of degree $m \geq 0$, the field $F_0 = pX$ satisfies

$$\operatorname{curl} F_0 = (m+2)F_0, \qquad \operatorname{div} F_0 = 0, \qquad F_0 \cdot F_0 = 0,$$

and therefore $F(\tau) = e^{-i(m+2)\tau}F_0$ is an exact source-free Maxwell solution. We prove the converse within the complex-line ansatz fX, identify the resulting $(m+1)$-dimensional holomorphic chiral subspace inside the full curl eigenspace of dimension $(m+1)(m+3)$, determine the nodal Hopf fibers, and compute the energy and magnetic helicity for an arbitrary polynomial. We then transport the modes conformally to Minkowski space, obtain the Bateman form $p(\alpha, \beta)\nabla\alpha \times \nabla\beta$, and give an explicit degree -4 Penrose representative whose contour transform reproduces the field with full normalization. The construction is a precise holomorphic sector of the known Maxwell spectrum rather than a classification of all Maxwell fields.

1 Introduction

Exact Maxwell fields adapted to the Hopf fibration arise naturally from several complementary viewpoints: Beltrami fields on the round three-sphere, Bateman variables in Minkowski space, conformal compactification, and the Penrose transform. Holomorphic polynomial constructions of null electromagnetic knots are well established in the literature [1, 2], while the curl spectrum of the round sphere is classical and can be expressed explicitly [3]. Conformal constructions from $\mathbb{R} \times S^3$ to Minkowski space have likewise been used to generate rational finite-energy Maxwell fields [4].

The purpose of this paper is to isolate one convention-explicit sector in which these descriptions can be matched exactly. The central construction is the embedding

$$\operatorname{Sym}^m(\mathbb{C}^2) \longrightarrow E^+_{m+2}, \qquad p \longmapsto p\,(X_2 + iX_3),$$

where E^+_{m+2} denotes the positive curl eigenspace with eigenvalue $m+2$ on the unit round S^3. The image has complex dimension $m+1$, so it is a distinguished chiral subspace rather than the full eigenspace. Within the fixed complex line generated by $X_2 + iX_3$, however, the construction is exhaustive.

We work first at unit radius and set $c = 1$. The radius and speed of light are restored by scaling in Corollary 2.3. Our Riemann–Silberstein convention is $F = E + iB$, with complex bilinear dot and cross products and Hermitian norm $F \cdot \overline{F}$. On the Einstein cylinder

$$g_{\mathrm{cyl}} = -\mathrm{d}\tau^2 + g_{S^3},$$

the source-free Maxwell equations take the form

$$\partial_\tau F = -i \operatorname{curl} F, \qquad \operatorname{div} F = 0. \tag{1.1}$$

The energy density is $u = (|E|^2 + |B|^2)/2$, and the Poynting vector is $S = E \times B$.

2 Hopf geometry and the exact Beltrami–Maxwell family

2.1 A global orthonormal frame

Write

$$S^3 = \{q \in \mathbb{R}^4 : |q| = 1\}, \qquad z_1 = q_1 + iq_2, \qquad z_2 = q_3 + iq_4.$$

Fix the global tangent frame

$$\begin{aligned} X_1 &= -q_2\partial_1 + q_1\partial_2 - q_4\partial_3 + q_3\partial_4, \\ X_2 &= -q_3\partial_1 + q_4\partial_2 + q_1\partial_3 - q_2\partial_4, \\ X_3 &= -q_4\partial_1 - q_3\partial_2 + q_2\partial_3 + q_1\partial_4, \end{aligned} \tag{2.1}$$

where $\partial_j = \partial/\partial q_j$. It is orthonormal for the unit round metric. We declare (X_1, X_2, X_3) positively oriented. Direct differentiation gives

$$[X_i, X_j] = -2\epsilon_{ijk} X_k. \tag{2.2}$$

If $\theta_j = X_j^\flat$ is the dual coframe, then

$$\mathrm{d}\theta_1 = 2\theta_2 \wedge \theta_3, \qquad \text{and cyclically,} \tag{2.3}$$

so

$$\operatorname{curl} X_j = 2X_j. \tag{2.4}$$

In particular θ_1 is a contact form and X_1 is its Reeb field.

Set

$$X := X_2 + iX_3, \qquad \theta := X^\flat = \theta_2 + i\theta_3. \tag{2.5}$$

The key Cauchy–Riemann-type identities are

$$Xz_1 = Xz_2 = 0, \qquad X_1 z_j = iz_j. \tag{2.6}$$

For every smooth complex-valued function f one also has the exact identity

$$\begin{aligned} \operatorname{curl}(fX) &= (2f - iX_1 f)X + i(Xf)X_1, \\ \operatorname{div}(fX) &= Xf. \end{aligned} \tag{2.7}$$

It follows by expanding $\nabla f \times X + f \operatorname{curl} X$ in the frame.

2.2 Main theorem

Theorem 2.1 (Holomorphic Hopf–Beltrami modes)**.** *Let $p(z_1, z_2)$ be a homogeneous holomorphic polynomial of degree $m \geq 0$. Then*

$$F_0 = p(z_1, z_2)X \tag{2.8}$$

is a globally smooth complex vector field on S^3 satisfying

$$\operatorname{curl} F_0 = (m+2)F_0, \qquad \operatorname{div} F_0 = 0, \qquad F_0 \cdot F_0 = 0. \tag{2.9}$$

Consequently

$$F(\tau) = e^{-i(m+2)\tau} F_0 \tag{2.10}$$

is a global source-free Maxwell field on $\mathbb{R} \times S^3$. If $F = E + iB$, then

$$|E|^2 = |B|^2 = |p|^2, \qquad E \cdot B = 0, \qquad S = |p|^2 X_1. \tag{2.11}$$

Proof. By (2.6), holomorphicity gives $Xp = 0$, while homogeneity gives

$$X_1 p = imp.$$

Equation (2.7) therefore yields

$$\operatorname{curl}(pX) = (2p - iX_1 p)X = (m+2)pX, \qquad \operatorname{div}(pX) = 0.$$

Since $X \cdot X = 1 - 1 = 0$, the field is null in the complex-bilinear sense. The time dependence (2.10) then satisfies (1.1).

To obtain the real fields, write $e^{-i(m+2)\tau} p = a + ib$. Then

$$E = aX_2 - bX_3, \qquad B = bX_2 + aX_3,$$

which immediately gives (2.11). □

The same construction has a simple representation-theoretic weight. From $[X_1, X] = 2iX$ and $X_1 p = imp$,

$$\mathcal{L}_{X_1} F_0 = i(m+2)F_0, \qquad (\partial_\tau + \mathcal{L}_{X_1})F = 0. \tag{2.12}$$

Thus the additional two units in the curl eigenvalue are carried by the complex polarization vector itself.

2.3 Converse inside the fixed complex line

Proposition 2.2 (Converse within the ansatz)**.** *Let f be smooth and suppose that the nonzero field fX is a curl eigenfield,*

$$\operatorname{curl}(fX) = \lambda f X.$$

Then f is the restriction to S^3 of a homogeneous holomorphic polynomial of some degree $m \geq 0$, and $\lambda = m + 2$.

Proof. Equation (2.7) gives

$$Xf = 0, \qquad X_1 f = i\mu f, \qquad \mu := \lambda - 2. \tag{2.13}$$

The Hopf flow generated by X_1 has period 2π; hence $e^{2\pi i\mu} = 1$, so $\mu \in \mathbb{Z}$.

The fields X_j are divergence free. In the Hermitian L^2 inner product, therefore, $X^* = -\overline{X}$. Using $[X, \overline{X}] = 4iX_1$ and $Xf = 0$,

$$\|\overline{X} f\|_{L^2}^2 = \langle f, -X\overline{X} f\rangle = 4\mu \|f\|_{L^2}^2. \tag{2.14}$$

Thus $\mu = m \geq 0$. For the nonpositive scalar Laplacian

$$\Delta = X_1^2 + X_2^2 + X_3^2$$

we have

$$\Delta f = -m(m+2)f. \tag{2.15}$$

Hence f lies in the degree-m scalar spherical-harmonic space, i.e. it is the restriction of a homogeneous harmonic polynomial of degree m on $\mathbb{R}^4$. Under the Hopf action, a monomial with a holomorphic and b antiholomorphic factors has total degree $a + b$ and weight $a - b$. The simultaneous conditions degree m and weight m force $b = 0$. Therefore the harmonic representative is holomorphic in z_1, z_2, proving the claim. □

Corollary 2.3 (Spectral position and radius scaling)**.** *For each $m \geq 0$, the fields pX form a complex vector space of dimension $m+1$. The full complexified positive curl eigenspace at eigenvalue $m+2$ on the unit round sphere has dimension*

$$\dim E^{+}_{m+2} = (m+1)(m+3), \tag{2.16}$$

so the holomorphic family is a proper subspace. On the round sphere of radius R, the eigenvalue and Maxwell frequency become

$$\lambda_m = \frac{m+2}{R}, \qquad \omega_m = \frac{c(m+2)}{R}. \tag{2.17}$$

Proof. The polynomial space $\mathrm{Sym}^m(\mathbb{C}^2)$ has dimension $m+1$. The full multiplicity formula (2.16) is the standard curl spectrum of the round three-sphere [3]. Rescaling the metric by R^2 rescales curl by R^{-1}, and restoring c in Maxwell's equation gives (2.17). □

2.4 Nodal Hopf fibers

A nonzero binary homogeneous polynomial factors over $\mathbb{C}$ into m linear factors, counted with multiplicity. The zero set of each factor is a complex line through the origin in $\mathbb{C}^2$, whose intersection with S^3 is a Hopf fiber. Hence, for $m > 0$, the nodal set of F_0 is a nonempty union of at most m distinct Hopf fibers. At those fibers the field and Poynting vector vanish. This nodal statement should not be confused with a classification of electric or magnetic field lines.

3 Energy and magnetic helicity

Let

$$p(z_1, z_2) = \sum_{a=0}^{m} c_a z_1^a z_2^{m-a}, \qquad \mathcal{N}(p) := \int_{S^3} |p|^2 \mathrm{d}V. \tag{3.1}$$

With Hopf coordinates

$$z_1 = \cos\eta\, e^{i\psi_1}, \qquad z_2 = \sin\eta\, e^{i\psi_2},$$

and volume form

$$\mathrm{d}V = \cos\eta \sin\eta\, \mathrm{d}\eta \mathrm{d}\psi_1 \mathrm{d}\psi_2,$$

phase integration makes distinct monomials orthogonal. Using the beta integral gives

$$\boxed{\mathcal{N}(p) = \frac{2\pi^2}{(m+1)!} \sum_{a=0}^{m} |c_a|^2 a!(m-a)!.} \tag{3.2}$$

Since B is a real Beltrami field with eigenvalue $m+2$, a global vector potential is

$$A_B = \frac{B}{m+2}.$$

The magnetic helicity on the closed sphere is therefore

$$\boxed{\mathcal{H}_B = \int_{S^3} A_B \cdot B\, \mathrm{d}V = \frac{\mathcal{N}(p)}{m+2}.} \tag{3.3}$$

The cylinder energy is

$$\mathcal{E}_{\mathrm{cyl}} = \int_{S^3} \frac{|E|^2 + |B|^2}{2}\, \mathrm{d}V = \mathcal{N}(p). \tag{3.4}$$

For the monomial $p = C z_1^m$,

$$\mathcal{H}_B = \frac{2\pi^2 |C|^2}{(m+1)(m+2)}. \tag{3.5}$$

The overall amplitude of p is continuous, so these formulas do not by themselves imply an integer helicity quantization.

4 Conformal transport and Bateman variables

Use Minkowski coordinates (t, x, y, z), let $r^2 = x^2 + y^2 + z^2$, and define

$$D = r^2 - (t - i)^2, \qquad s = |D|,$$
$$\tau = \arctan(t + r) + \arctan(t - r), \qquad \Omega = \frac{2}{s}. \tag{4.1}$$

The corresponding point of the unit three-sphere is

$$(q_1, q_2, q_3, q_4) = \frac{1}{s}(r^2 - t^2 - 1,\ 2z,\ 2x,\ -2y), \tag{4.2}$$

with $D = se^{i\tau}$ and

$$\Phi^* g_{\text{cyl}} = \Omega^2 \eta, \qquad \eta = -\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2. \tag{4.3}$$

Define

$$\alpha = e^{-i\tau} z_1 = \frac{r^2 - t^2 - 1 + 2iz}{D}, \qquad \beta = e^{-i\tau} z_2 = \frac{2(x - iy)}{D}. \tag{4.4}$$

For real spacetime points $D \neq 0$, and one has $|\alpha|^2 + |\beta|^2 = 1$.

The sphere identities

$$z_1 \mathrm{d}z_2 - z_2 \mathrm{d}z_1 = \theta, \qquad \mathrm{d}z_1 \wedge \mathrm{d}z_2 = *_3\theta \tag{4.5}$$

imply

$$\mathrm{d}\alpha \wedge \mathrm{d}\beta = e^{-2i\tau}(*_3\theta - i\mathrm{d}\tau \wedge \theta). \tag{4.6}$$

For a complex spatial field F, write its associated spacetime two-form as

$$\mathcal{F}[F] = *_3 F^\flat - i\mathrm{d}\tau \wedge F^\flat. \tag{4.7}$$

Because the Hodge star on two-forms is conformally invariant in four dimensions, multiplication of (4.6) by

$$p(\alpha, \beta) = e^{-im\tau} p(z_1, z_2)$$

shows that the Minkowski image of the cylinder solution is

$$\boxed{F_{\text{M}} = p(\alpha, \beta)\, \nabla\alpha \times \nabla\beta.} \tag{4.8}$$

This is precisely the Bateman construction for this sector [1].

For explicit computations, put

$$u = x - iy, \qquad q = t - z - i. \tag{4.9}$$

Then

$$G := \nabla\alpha \times \nabla\beta = \frac{4}{D^3} \begin{pmatrix} q^2 - u^2 \\ -i(q^2 + u^2) \\ 2qu \end{pmatrix}. \tag{4.10}$$

Direct differentiation gives

$$G = i(\partial_t \alpha\, \nabla\beta - \partial_t \beta\, \nabla\alpha), \tag{4.11}$$

so (4.8) satisfies the full vacuum Maxwell equations and remains null. Since $p(\alpha, \beta)$ is bounded on real spacetime and $G = O(r^{-4})$ at fixed time, the resulting Minkowski fields are smooth and have finite energy on every constant-time slice.

5 Explicit Penrose representative

5.1 Twistor conventions and cohomology class

Let

$$Z = (\omega^0, \omega^1, \pi_{0'}, \pi_{1'}), \qquad x^{AA'} = \begin{pmatrix} t+z & x+iy \\ x-iy & t-z \end{pmatrix}, \tag{5.1}$$

with incidence relation

$$\omega^A = i x^{AA'} \pi_{A'}. \tag{5.2}$$

We use

$$\Delta\pi = \pi_{0'} \mathrm{d}\pi_{1'} - \pi_{1'} \mathrm{d}\pi_{0'} \tag{5.3}$$

and identify a symmetric Maxwell spinor with the Riemann–Silberstein vector by

$$F = (\phi_{0'0'} - \phi_{1'1'}, -i(\phi_{0'0'} + \phi_{1'1'}), -2\phi_{0'1'}). \tag{5.4}$$

These conventions fix the normalization and duality signs used below. Standard Penrose-transform treatments of massless fields may be found in [5, 2].

Define four linear forms on twistor space,

$$\begin{aligned} \mathsf{A} &= \omega^1 + \pi_{1'}, & \mathsf{B} &= \omega^0 + \pi_{0'}, \\ \mathsf{C} &= \omega^0 - \pi_{0'}, & \mathsf{W} &= -i(\omega^1 - \pi_{1'}). \end{aligned} \tag{5.5}$$

Let

$$\mathcal{U} = \mathbb{CP}^3 \setminus \{\mathsf{A} = \mathsf{B} = 0\} = U_A \cup U_B, \tag{5.6}$$

where $U_A = \{\mathsf{A} \neq 0\}$ and $U_B = \{\mathsf{B} \neq 0\}$. On the overlap define

$$\boxed{f_p(Z) = -4 \frac{p(\mathsf{C}, \mathsf{W})}{\mathsf{A}\, \mathsf{B}^{m+3}}.} \tag{5.7}$$

The numerator has degree m, so f_p is homogeneous of degree -4 and defines a Čech representative in

$$H^1(\mathcal{U}, \mathcal{O}(-4)).$$

The deleted projective line is the twistor line of the complex point $(t, x, y, z) = (i, 0, 0, 0)$. Real Minkowski twistor lines do not meet it because the determinant of the restrictions of A and B is $-D \neq 0$.

Theorem 5.1 (Penrose transform)**.** *For every homogeneous holomorphic polynomial p of degree m, the Penrose contour integral of (5.7), enclosing the simple A-pole and excluding the B-pole, is*

$$\phi_{A'B'} = \frac{4p(\alpha, \beta)}{D^3} \begin{pmatrix} q^2 & -qu \\ -qu & u^2 \end{pmatrix}_{A'B'}. \tag{5.8}$$

Under (5.4), this is exactly the Minkowski Maxwell field (4.8). Where nonzero, the Maxwell spinor has rank one.

Proof. In the affine chart $\pi = (1, \zeta)$, set

$$v = x + iy, \qquad s_- = t + z - i.$$

The incidence relation gives

$$\mathsf{A} = i(u + q\zeta), \qquad \mathsf{B} = i(s_- + v\zeta). \tag{5.9}$$

For real spacetime points, $q = t - z - i \neq 0$, and the enclosed pole is

$$\zeta_0 = -\frac{u}{q}.$$

At this pole,

$$\mathsf{B}(\zeta_0) = -\frac{iD}{q}, \qquad \frac{\mathsf{C}(\zeta_0)}{\mathsf{B}(\zeta_0)} = \alpha, \qquad \frac{\mathsf{W}(\zeta_0)}{\mathsf{B}(\zeta_0)} = \beta. \tag{5.10}$$

By homogeneity,

$$p(\mathsf{C}, \mathsf{W}) = \mathsf{B}^m p(\alpha, \beta)$$

at the pole. Since $\Delta\pi = \mathrm{d}\zeta$ in this chart, the residue is

$$\begin{aligned} \phi_{A'B'} &= \frac{1}{2\pi i} \oint \pi_{A'} \pi_{B'} f_p(Z(x, \pi))\, \Delta\pi \\ &= \frac{-4p(\alpha, \beta)}{(iq)(-iD/q)^3} (1, -u/q)_{A'} (1, -u/q)_{B'}, \end{aligned} \tag{5.11}$$

which simplifies to (5.8). Substitution into (5.4) yields

$$p(\alpha, \beta) \frac{4}{D^3} (q^2 - u^2, -i(q^2 + u^2), 2qu),$$

namely (4.8) with (4.10). □

5.2 Cylinder-time weight

In the adapted twistor coordinates (5.5), define

$$\mathcal{T} = \frac{i}{2} \left(\mathsf{A}\partial_{\mathsf{A}} + \mathsf{B}\partial_{\mathsf{B}} - \mathsf{C}\partial_{\mathsf{C}} - \mathsf{W}\partial_{\mathsf{W}} \right). \tag{5.12}$$

This vector field is the twistor lift of the conformal Killing field corresponding to cylinder-time translation. In Minkowski coordinates the latter is

$$K = \frac{1}{2}(1 + t^2 + r^2)\partial_t + t(x\partial_x + y\partial_y + z\partial_z), \tag{5.13}$$

which is mapped by (4.2) to ∂_τ.

Since p has degree m in (C, W), applying (5.12) to (5.7) gives

$$\mathcal{T} f_p = \frac{i}{2} [-1 - (m + 3) - m] f_p = -i(m + 2) f_p. \tag{5.14}$$

Thus the same frequency $m + 2$ appears as the conformal-generator weight of the twistor representative and as the curl eigenvalue of the cylinder field. In the representative (5.7), the two displayed pole orders are 1 and $m + 3$; the frequency is therefore a generator weight, not simply a pole order.

6 Conclusion

For every $m \geq 0$, homogeneous holomorphic polynomials of degree m define a canonical $(m + 1)$-dimensional chiral subspace of the positive curl eigenspace on the round three-sphere through

$$p \longmapsto p(X_2 + iX_3).$$

These fields are globally smooth, divergence free, Beltrami, and null. Their exact cylinder evolution has frequency $m + 2$, their Poynting flux follows the Hopf direction away from nodal

fibers, and their energy and magnetic helicity are given explicitly by (3.2)–(3.4). Within the fixed complex-line ansatz, Proposition 2.2 shows that no other smooth curl eigenfields occur.

Conformal transport gives the Minkowski Bateman field (4.8), and Theorem 5.1 supplies an explicit degree -4 twistor representative with full normalization. These results place the construction simultaneously in the Hopf/Beltrami, Maxwell, Bateman, and Penrose descriptions while keeping its spectral scope precise: it is an explicit holomorphic sector of the Maxwell field space, not a complete classification.

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